# GEOMETRIC REGULARITY IS UNCHANGED BY A FINITELY GENERATED EXTENSION OF THE BASE FIELD

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is complete as written and ends with the pointwise criterion; no step is referred to the reader. The author's parenthetical that the point over x is unique for a purely inseparable extension is reproduced verbatim.

*Editorial note.* The counted spans carry 3 navigation stubs and no bibliography citation.

*Editorial note.* The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

*Editorial note.* The label lemma-geometrically-regular-upstairs is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

*Allowed standard prerequisites (editorial).*

- Flat local homomorphisms of Noetherian local rings are faithfully flat, and regularity descends along faithfully flat maps. Algebra, Lemma algebra-lemma-flat-under-regular, invoked at proof lines 2497-2501. Supplies that regularity upstairs forces regularity downstairs.
- Existence of a common finitely generated extension of two given extensions of a field. Fields, Lemma fields-lemma-common-extension-field, invoked at proof lines 2489-2492. Supplies the field k"' and the point lying over both x' and x".
- Pointwise criterion for geometric regularity. Same chapter, Lemma lemma-geometrically-regular-at-point, invoked at proof lines 2502-2503. Supplies the passage from finite purely inseparable base changes to geometric regularity over k.

*Source reference.* Original lemma 00OF: sources/stacks-chow/tags.txt, source lines 893–893. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0H7K: sources/stacks-chow/tags.txt, source lines 21460–21460. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 038U: sources/stacks-varieties/pinned-original-source.tex, source lines 2382–2382. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $k$ be a field. Let $X$ be a scheme over $k$. Let $k'/k$ be a finitely generated field extension. Let $x \in X$ be a point, and let $x' \in X_{k'}$ be a point lying over $x$. The following are equivalent*

(1) *$X$ is geometrically regular at $x$,*
(2) *$X_{k'}$ is geometrically regular at $x'$.*

*In particular, $X$ is geometrically regular over $k$ if and only if $X_{k'}$ is geometrically regular over $k'$.*

*Proof.* It is clear that (1) implies (2). Assume (2). Let $k''/k$ be a finite purely inseparable field extension and let $x'' \in X_{k''}$ be a point lying over $x$ (actually it is unique). We can find a common, finitely generated, field extension $k'''/k$ of $k'/k$ and $k''/k$, see Fields, Lemma 0H7K and its proof. and a point $x''' \in X_{k'''}$ lying over both $x'$ and $x''$. Consider the map of local rings

$$\mathcal{O}_{X_{k''},x''} \longrightarrow \mathcal{O}_{X_{k'''},x'''}.$$

This is a flat local ring homomorphism of Noetherian local rings and hence faithfully flat. By (2) we see that the local ring on the right is regular. Thus by Algebra, Lemma 00OF we conclude that $\mathcal{O}_{X_{k''},x''}$ is regular. By Lemma 038U we see that $X$ is geometrically regular at $x$. □